MadhuTwin: system architecture

A hybrid digital twin for Type 2 Diabetes in India: a personalised physiology model fused with machine learning to predict glucose excursions 2 hours ahead

1 · DATA SOURCES

Static · EHR

Synthea CSV schema (FHIR-mappable)

- Demographics, region, urban/rural
- Diagnoses (SNOMED CT)
- Medications (RxNorm)
- Labs (LOINC): HbA1c trend, FPG, insulin, lipids, eGFR, UACR
- Genotype (record only; no gain)

Dynamic · wearables

5-minute streams, 21 days/patient

- CGM glucose (Libre/Dexcom-like)
- Heart rate, HRV (RMSSD)
- Steps / activity
- Sleep stages (light/deep/REM)
- Food log app: meal, carbs, GI (incomplete & under-estimated, as in real life)

Apple Health / Google Fit style export

Virtual-patient simulator

Stands in for real patients (DPDP/HIPAA)

- 300 India-context synthetic adults
- 37 regional foods, festivals, late dinners, unlogged chai
- Sleep, stress, illness, missed doses, sensor noise
- Hidden ground truth kept for validation only

2 · INGEST & HARMONISE

EHR loader

- Reads Synthea/FHIR codes
- Latest labs + 12-month trend
- HOMA-IR, HOMA-B, TG/HDL
- 44 static features per patient

madhutwin/ehr.py

Stream harmoniser

- Align to a 5-minute grid
- Gap-aware: sensor warm-up, dropouts, watch off-wrist
- Night windows 18:00→12:00
- Personal baselines (HR, HRV)
- Meal log → gut inputs

twin.load_arrays · features.py

3 · LAYER 1 — DIGITAL TWIN (state)

Physiology twin

extended Bergman minimal model, per patient

A · Personalise

Fit SI (insulin sensitivity), σ (beta-cell gain), Gb, dawn rise on week-1 CGM + meals + steps with an EHR-informed prior (HOMA-IR → SI, HOMA-B → σ)

regularised grid fit (146k sims)

B · Observe (every 5 min)

State estimator nudged by CGM: glucose, insulin action, carbs still in the gut, plasma ahead of lagging sensor

closed-loop observer

C · Project 2 h ahead

(i) food already eaten
(ii) + the patient's habitual next meal learned from logs

→ projected peak, +30/60/120

1.8M projections in ~6 s

D · What-if engine

Counterfactuals: food swap, portion size, post-meal walk

Same equations in Python and in the browser (JS)

Layer 1: what state is this person in?

mechanistic · personal · simulates interventions

4 · LAYER 2 — AI PREDICTION

Fusion feature store

107 leak-free features per 5-min step

- CGM dynamics & personal history (17)
- Meals, steps, HR, HRV, sleep (27)
- Twin states & projections (16)
- EHR labs, Dx, Rx, trends (44)
- Clock / weekday (3)

features.py

ML layer (LightGBM)

- 2-h risk of excursion ≥180
- 2-h risk of excursion ≥250
- Quantile forecasts +30/+60/+120 (10th, 50th, 90th percentile)
- Threshold from validation F1; debounced alerts (1 per hour)
- SHAP → plain-language reasons
- Sick-day detector: a week of HR / HRV / sleep vs own baseline

Patient-level splits; conformal 80% bands

Validation harness

- AUROC/AUPRC + 95% CIs, Clarke grid
- Delivered-alert lead time, false alerts
- Twin vs hidden physiology; what-if
- Ablation; model-mismatch stress test
- **Open cohorts: 265 (retrospective)** CGM + labs only, re-trained, CV

Layer 2: will an event happen, when, why?

learned from 210 patients · calibrated · explained

5 · CLINICIAN CONSOLE

Doctor's console

static web app · runs offline

Triage list

Ranked by risk × severity × confidence × time-to-event
"Replay live" twin clock

Now view

Risk ± confidence, time to event
Twin state vs personal baseline
Why (reasons in %) → action
CGM + forecast cone, alert log

What-if simulator

Swap rice → millet, portion, post-meal walk → peak, time >180, event probability

7-day AGP

TIR/TAR/TBR, GMI, CV vs International Consensus targets

Record & physiology

Labs, Rx, genotype, twin SI/σ, disposition index, meal habits

Patient app · sick-day · model card

Nudges in English / हिन्दी; sick-day flag; open-cohort + subgroup results

Every 5 minutes, per patient

CGM reading arrives → twin observer updates hidden state → 2-h projections → 107 fused features → LightGBM risk + forecasts → debounced alert + SHAP reasons → console / nudge

Outputs: doctor console · patient app nudges (English / हिन्दी) · FHIR R4 sample bundle (ABDM: proposed path) · REST API (FastAPI, Docker) · CI-tested · replayed data, not live devices

Privacy by design: synthetic patients + open de-identified cohorts only. Production path: ABDM / FHIR R4, Health Connect / HealthKit, on-premise inference, consent under the DPDP Act 2023.

Inside the twin and the evaluation

Equations, components and how each claim is tested

Physiology model (per patient, 1-minute or 5-minute steps)

STATE	EQUATION	MEANING
G	$dG/dt = -(S_G + X + k_{ex}\cdot steps)\cdot G + S_G\cdot G_b(t) + R_a/V - renal(G)$	plasma glucose
X	$dX/dt = -p_2\cdot X + p_2\cdot S_I(t)\cdot I$	insulin action
I	$dI/dt = -n\cdot I + Y$	insulin above basal
Y	$dY/dt = \alpha\cdot(\sigma\cdot\max(G - h, 0) - Y)$	delayed beta-cell secretion
F, S	two-compartment fast / slow gut chains; split by GI	meal absorption R_a
IG	$dIG/dt = (G - IG)/\tau$, τ = 8 min	what the CGM sees

$S_I(t)$ carries a circadian rhythm and a post-exercise boost; $G_b(t)$ carries the dawn phenomenon. SGLT2 inhibitors lower the renal threshold. In the simulator (not the twin), sleep debt, stress, illness and missed doses also modulate S_I and G_b; the ML layer learns those from HRV, heart rate and sleep.

Technology stack

Language / numerics	Python 3.11, NumPy, pandas, SciPy (vectorised ODEs; no GPU needed)
ML	LightGBM 4 (binary + quantile), scikit-learn metrics, native TreeSHAP
Data formats	Synthea CSV (SNOMED CT, LOINC, RxNorm codes), Parquet time series
Console	Dependency-free HTML/CSS/JS + SVG; twin ported to JavaScript
Reproducibility	`python run.py` rebuilds models → figures → console; seed 2026; versions pinned in requirements-lock.txt
Interop & deploy	FHIR R4 Bundle (RiskAssessment, Observation); FastAPI REST service; Dockerfile; GitHub Actions CI
Testing	pytest (28 tests): physiology sanity, leakage guards, JS/Python twin parity, Clarke grid, conformal coverage, FHIR, API

How each claim is tested

1. **Prediction (held-out synthetic).** 45 unseen virtual patients, days 8–21, every 5 min. AUROC, AUPRC, Brier, reliability curve.
2. **Alert usefulness.** One alert policy (≤1/hour, ≥9 of 12 readings present) shared with the console. For every excursion: was a *delivered* alert raised in the prior 2 h (one-to-one matching), and how early? False alerts and precision from the same alerts.
3. **Fusion value.** Ablation: EHR only, wearables only, fusion, fusion + twin; plus a cold-start test for the first 48 h of a new patient.
4. **Twin fidelity (synthetic).** Each virtual patient's hidden physiology is known; we report rank correlations with the truth. Simulator and Twin share equations, so this is an upper bound; a **model-mismatch stress test** changes the generator's physics.
5. **What-if accuracy (synthetic).** Interventions are re-run on the hidden virtual patient and compared with the Twin. Under the stress test, rankings held but effect sizes were biased; real-patient effects are unvalidated.
6. **Retrospective open cohorts.** 265 adults (Hall 2018; Colas 2019; only 10 with T2D), CGM + labs only. Pipeline re-trained with patient-level CV, target ≥140 mg/dL: CGM-only and CGM + EHR reported separately, plus SSPG insulin-resistance checks.
7. **Leakage guards.** Patient-level split (210 train / 45 validation / 45 test; no person in two sets) *and* temporal split (twin fitted on days 1–7, evaluated on days 8–21). Every feature uses data at or before the decision time; personal baselines use completed nights only; habits come from week 1.

India-context design choices (synthetic simulation)

1. Asian BMI and waist cut-offs. (Genotype is in the record but adds no predictive value, so the risk model excludes it.)
2. 37-item regional food library (idli, poha, rajma chawal, fish-rice, millets) with carbs and GI.
3. Late dinners, sweet chai, festival thalis, office commutes, weekend drift.
4. Metformin, sulfonylurea, DPP-4i, SGLT2i and basal insulin mix typical of Indian practice (in the record; SGLT2i what-if kept as a research option, not in the demo).
5. Built to run on modest hardware and offline, for tier-2/3 clinics.

Headline results · held-out synthetic evaluation (45 virtual patients) + retrospective open-cohort check

0.912 AUROC, 2-h excursion ≥180 mg/dL	**75%** of spikes warned ≥30 min ahead (delivered alerts)	**75 min** median warning, alert → onset	**2.3** false alerts per patient-day	**21.0** mg/dL error, 2-h forecast (vs 34.5 persistence)	**0.78** rank corr. twin vs true insulin sensitivity	**0.93** corr. predicted vs true rice→millet effect (synthetic truth)	**0.809 / 0.793** AUROC CGM-only / CGM+EHR, 265 open-cohort adults (≥140, retrospective CV)	**0.97** sick-day detection AUROC (HR/HRV/sleep)

Why a twin, and how it would be deployed

What the twin adds over ordinary AI · the patient state vector · prototype vs production · privacy and governance

Ordinary AI vs digital twin

	ORDINARY AI RISK MODEL	MADHUTWIN
Question answered	“Will glucose go high?”	“What state is this person in, what happens next, and what if we act?”
Patient model	Shared weights, no per-person state	Per-person physiology (SI, σ, Gb, dawn) + hidden state every 5 min
Data needed	Lots of labelled history	Physiology prior + EHR; useful from day 1 (twin matures over week 1)
Interventions	Cannot simulate	Food swap, portion, walk: simulated estimates, checked against synthetic ground truth
Explanation	Feature attributions	Attributions *and* mechanism (carbs in gut, insulin action)
Drift	Retrain	Nightly re-personalisation of SI and basal glucose

Measured value of the twin layer

WHERE	NO TWIN	WITH TWIN	GAIN
5 training patients	0.881	0.890	**+0.009**
10 training patients	0.891	0.896	**+0.005**
20 training patients	0.896	0.899	**+0.003**
40 training patients	0.900	0.901	**+0.001**
80 training patients	0.907	0.908	**+0.001**
Full data (210 patients)	0.911	0.912	+0.0007 [-0.0004, +0.0020]
False alerts/day at equal sensitivity	2.23	2.30	
Insulin-sensitivity estimate (ρ vs truth)	0.13 CGM only	0.78	fused
What-if: rice→millet / portion / walk (ρ)	0.93 / 0.91 / 0.68		synthetic truth

Honest reading: with 210 training patients the ML layer already learns most of what the twin knows, so the AUROC gain is within noise. The twin earns its place where data are scarce (new clinics, new patients), in estimating physiology, and in answering “what if?” questions, which a pure prediction model cannot answer at all.

Patient state vector (updated every 5 min)

STATE	SOURCE	UPDATED
Glucose (plasma, ahead of sensor)	CGM + observer	5 min
Carbs still in gut (fast / slow)	Meal log + twin	5 min
Insulin action X, insulin I	Twin	5 min
Activity, HR vs own baseline	Wearable	5 min
Sleep debt, HRV vs own baseline	Wearable	nightly
Insulin sensitivity SI (× baseline)	Twin re-fit	nightly
Basal glucose Gb	Twin re-fit	nightly
β-cell gain σ, dawn rise	Week-1 fit + EHR prior	weekly
HbA1c, eGFR, meds, comorbidity	EHR	on new result
Habitual next meal (time, carbs, GI)	Meal log	weekly

Prototype vs production

	THIS PROTOTYPE	PRODUCTION PATH
Data in	Synthetic CSV/Parquet; open CGM cohorts	ABDM / FHIR R4 from HMIS; Health Connect / HealthKit; CGM vendor APIs
Compute	Laptop CPU, no GPU	On-premise or Indian-region cloud; one container per clinic
Serving	FastAPI + static console	Same API behind hospital SSO; FHIR RiskAssessment written back to EHR
Models	Retrained by script	Versioned registry, drift monitor, shadow mode before go-live
Status	Research prototype	Prospective study; CDSCO SaMD pathway

Privacy and governance layer

1. **No real patient data in this repo.** Synthetic cohort + CC-BY de-identified open cohorts only.
2. **Consent (DPDP Act 2023).** Purpose-bound consent per data stream (CGM, watch, food log, EHR); revocable from the patient app; ABDM consent artefacts.
3. **Minimisation.** The twin needs 5-min aggregates, not raw accelerometry; no location, no free text leaves the device.
4. **Security.** Encryption at rest and in transit; role-based access; audit log of every view and alert acknowledgement.
5. **Human in the loop.** Advisory only: alerts go to the care team; no automatic dosing; every suggestion shows its reasons and confidence.
6. **Fairness monitoring (proposed).** AUROC with CIs by sex, age, BMI, HbA1c, region and medication; a release gate if any subgroup drops is a proposed governance step.
7. **Model card** (docs/model_card.md): intended use, out-of-scope use, data, metrics, limits.

Runs on modest hardware

Twin state update (1 patient)	`0.032 ms`
2-h twin projection	`0.82 ms`
Risk + SHAP reasons	`6.78 ms`
End-to-end alert (per patient, per 5 min)	`8.4 ms`
What-if 5-h simulation	`14.5 ms`
REST API risk request	`16.8 ms`
All model files	`25.7 MB`
Hardware	`2-core CPU, no GPU`

One CPU core can refresh about 35,502 patients every 5 minutes.